CEFET/RJ

GRÁFICOS, FIGURAS E TABELAS

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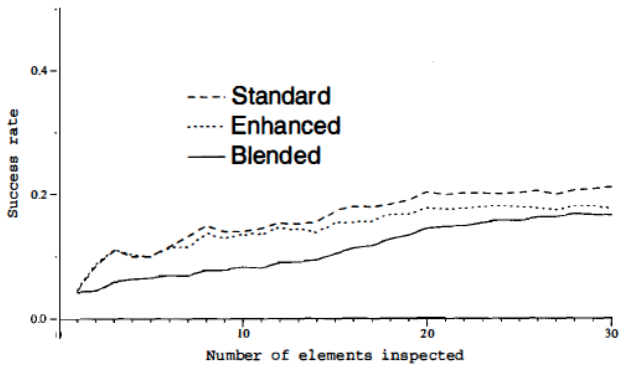
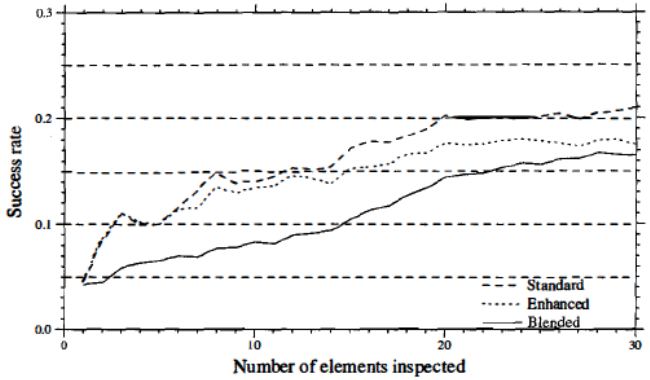
Gráficos

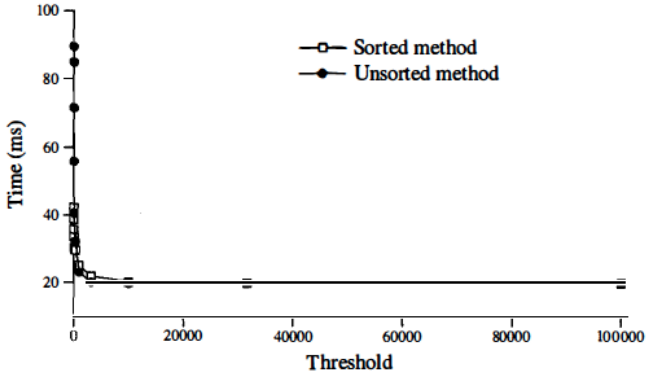
- Geralmente é a melhor forma de apresentar resultados numéricos
- Os gráficos devem ser usados para apresentar novas informações
  - Devem ser usados para apoiar uma hipótese
  - Cada gráfico deve trazer uma contribuição clara e uma perspectiva diferenciada em relação aos demais gráficos
    - Evite inundar o artigo com estatísticas “parecidas”, mesmo na forma gráfica, para não ser repetitivo

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Quais destes gráficos estão bem-feitos?

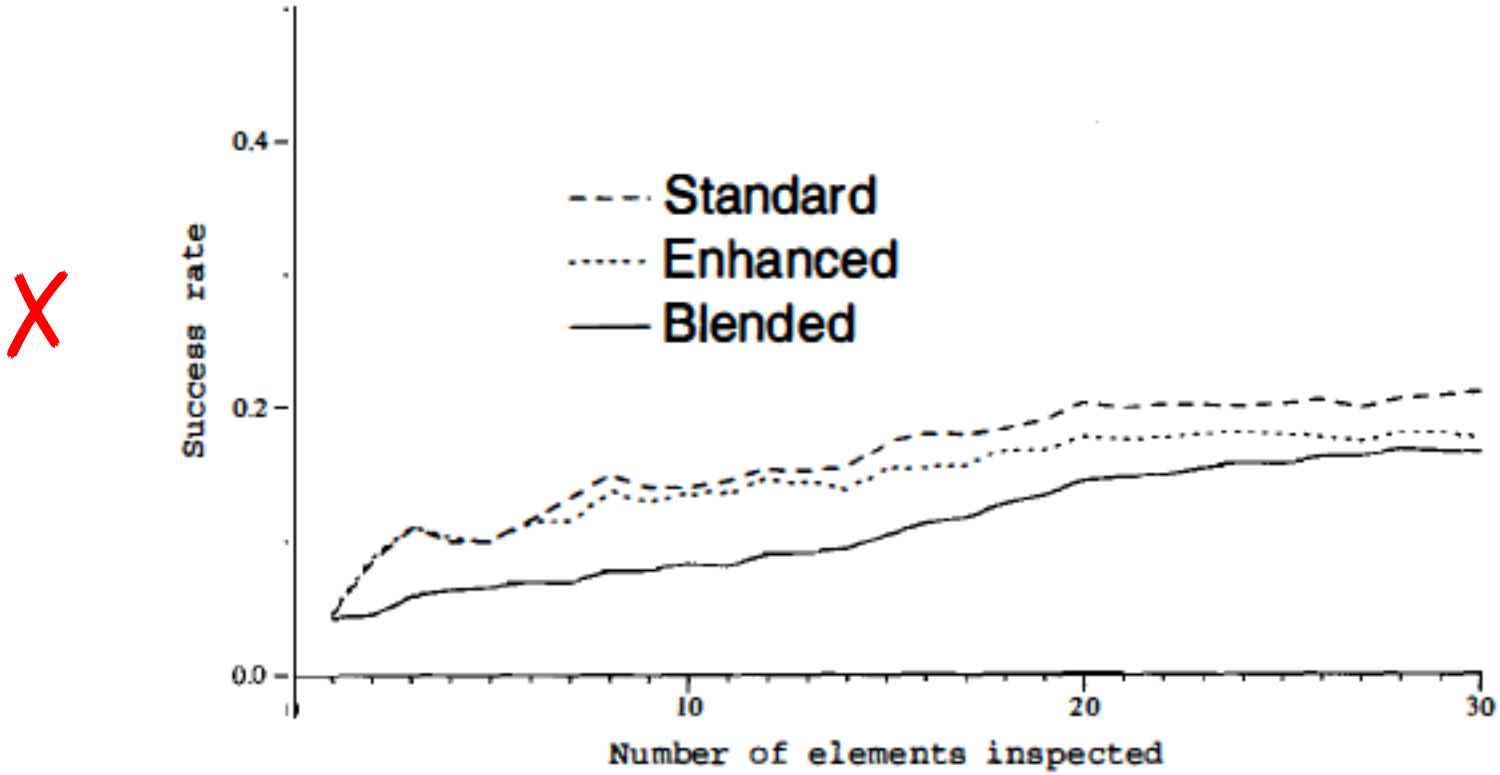






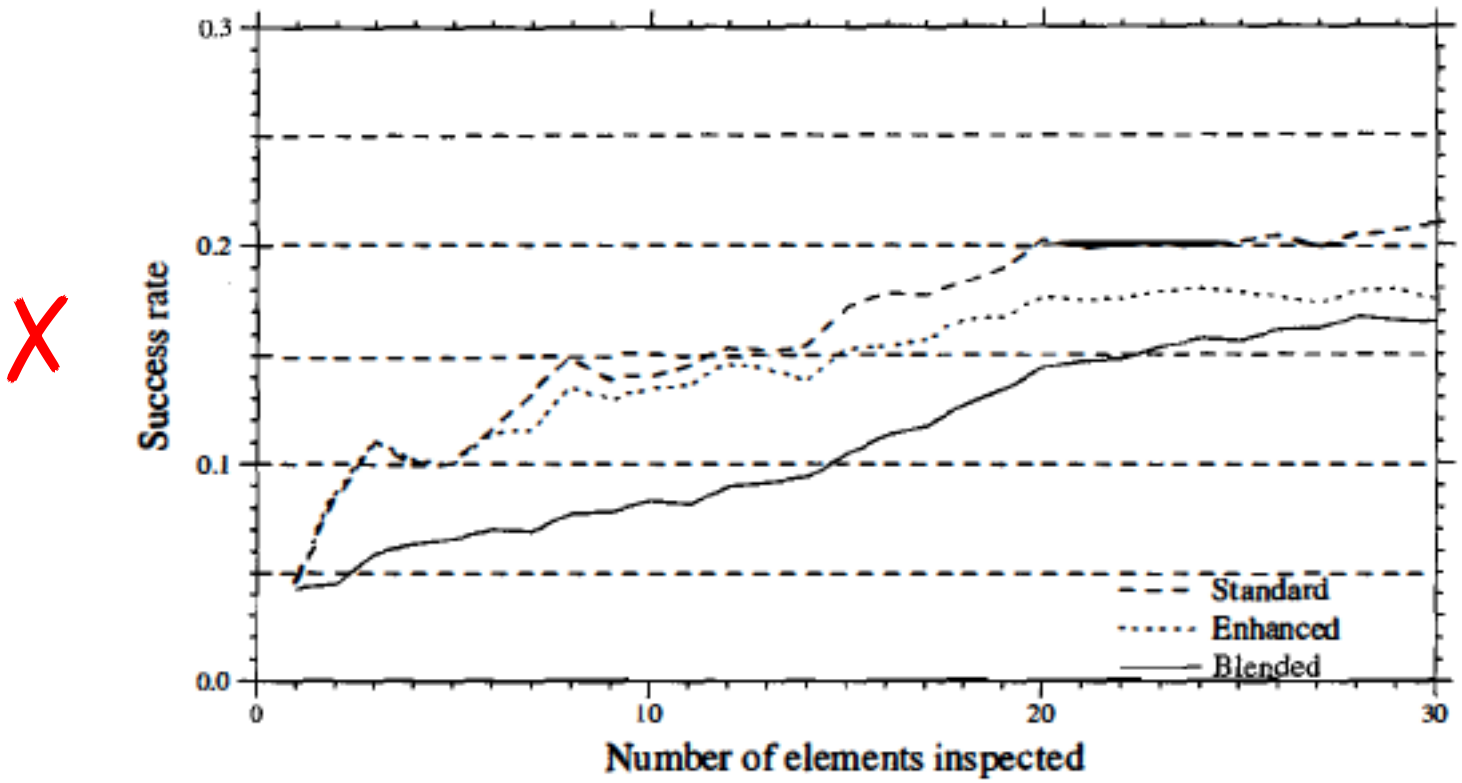
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Gráfico que não utiliza o eixo y por completo e a legenda posicionada em lugar inapropriado



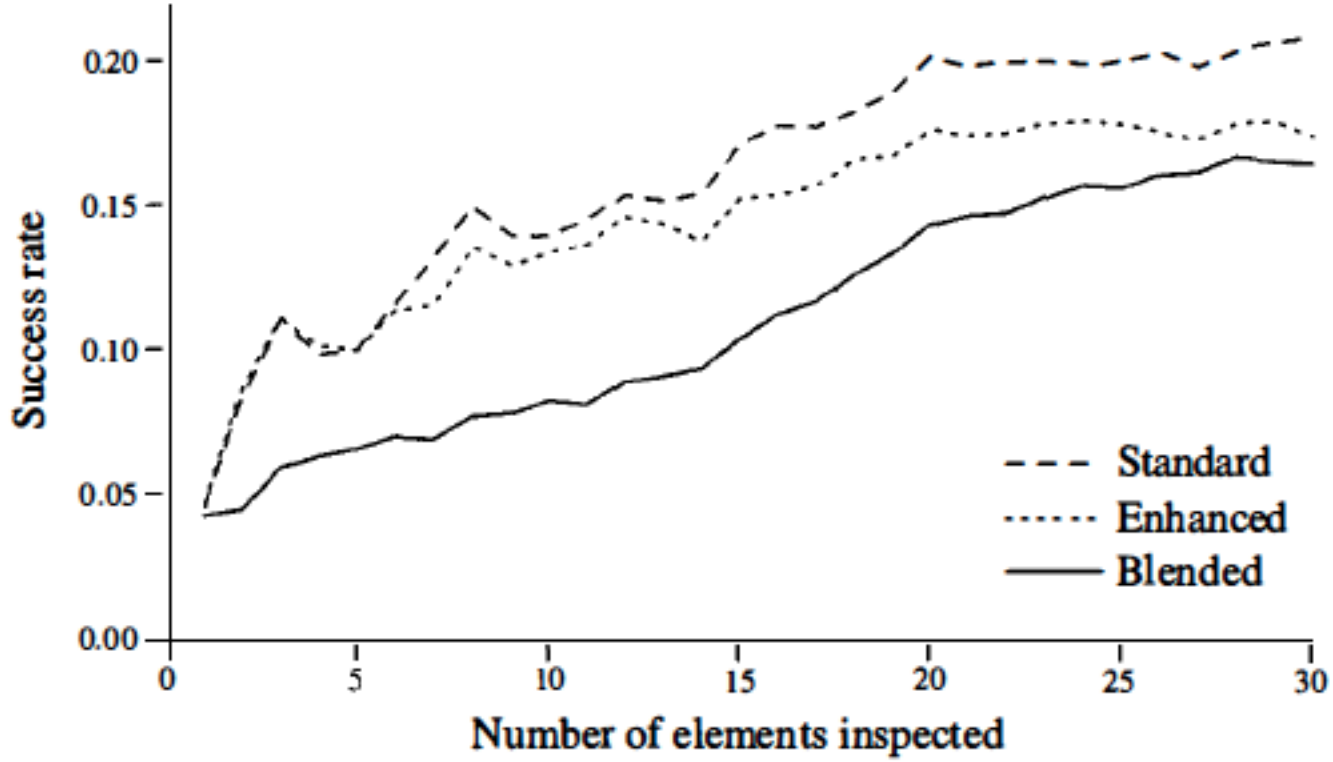
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Gráfico com excesso de ornamentos: as linhas de grade atrapalham a visualização dos dados



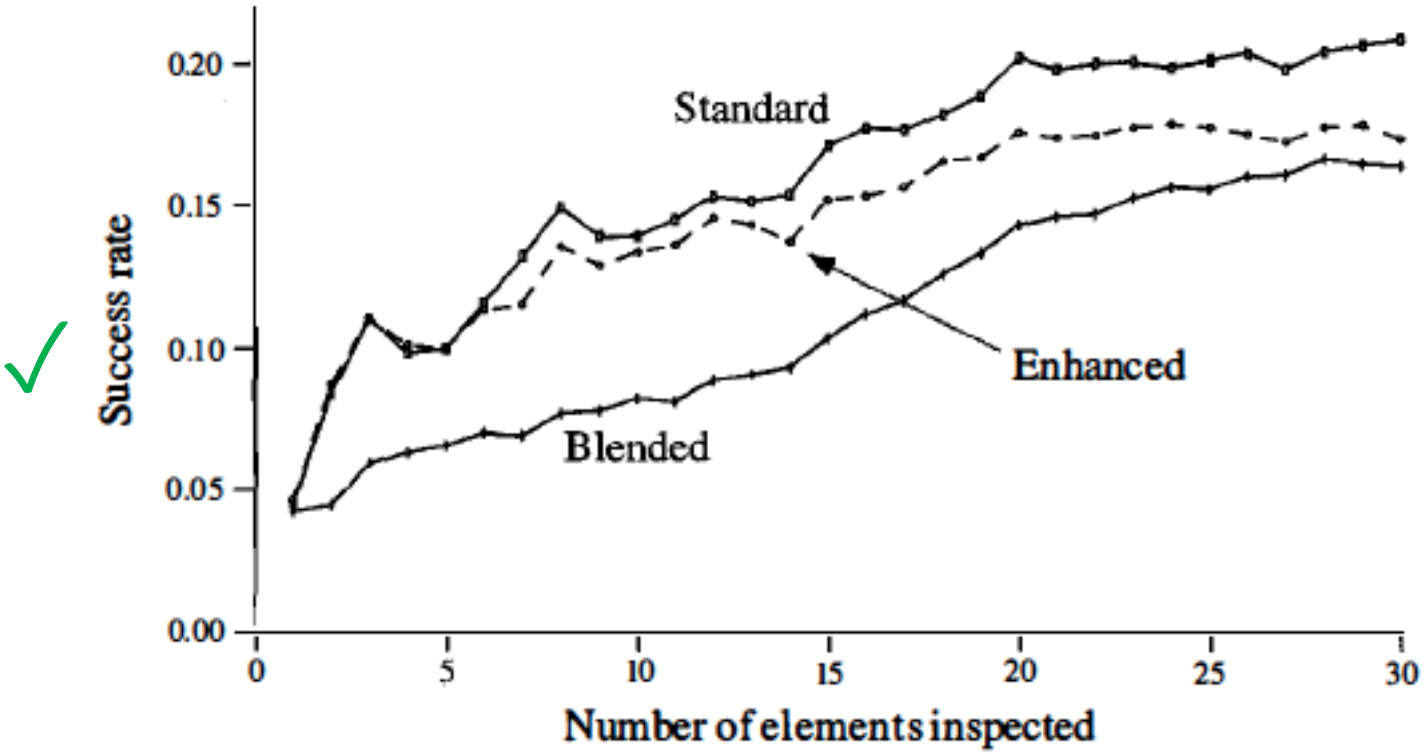
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Uso adequado do eixo y e legenda em posição adequada



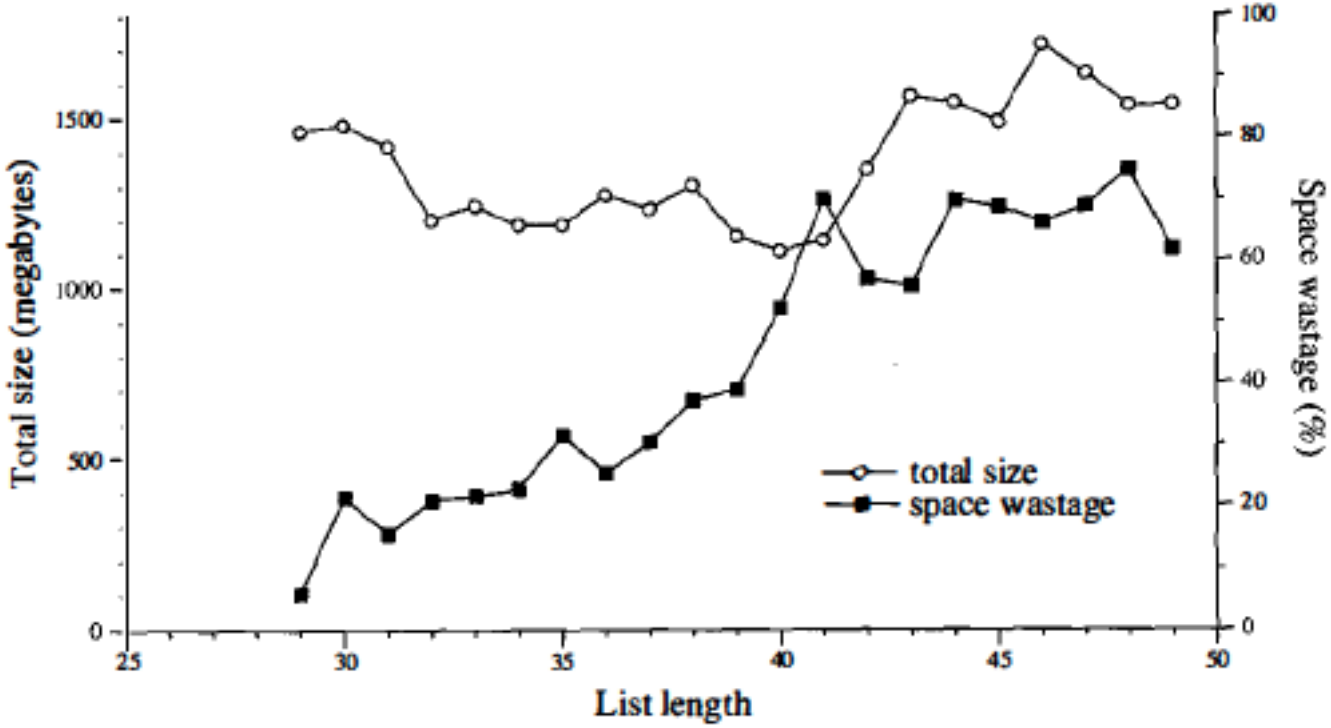
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Marcações incluídas para indicar que os dados são discretos



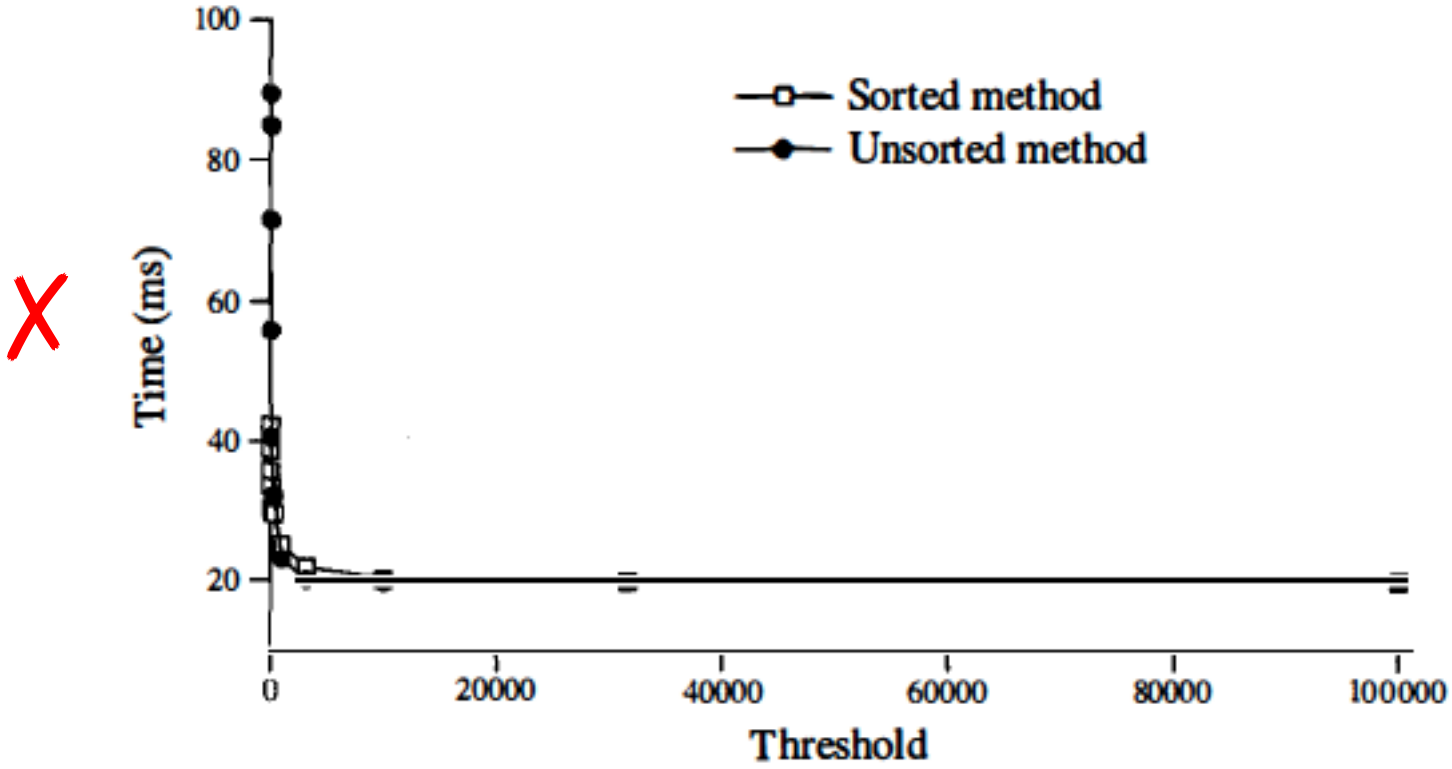
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Gráfico com duas variáveis dependentes



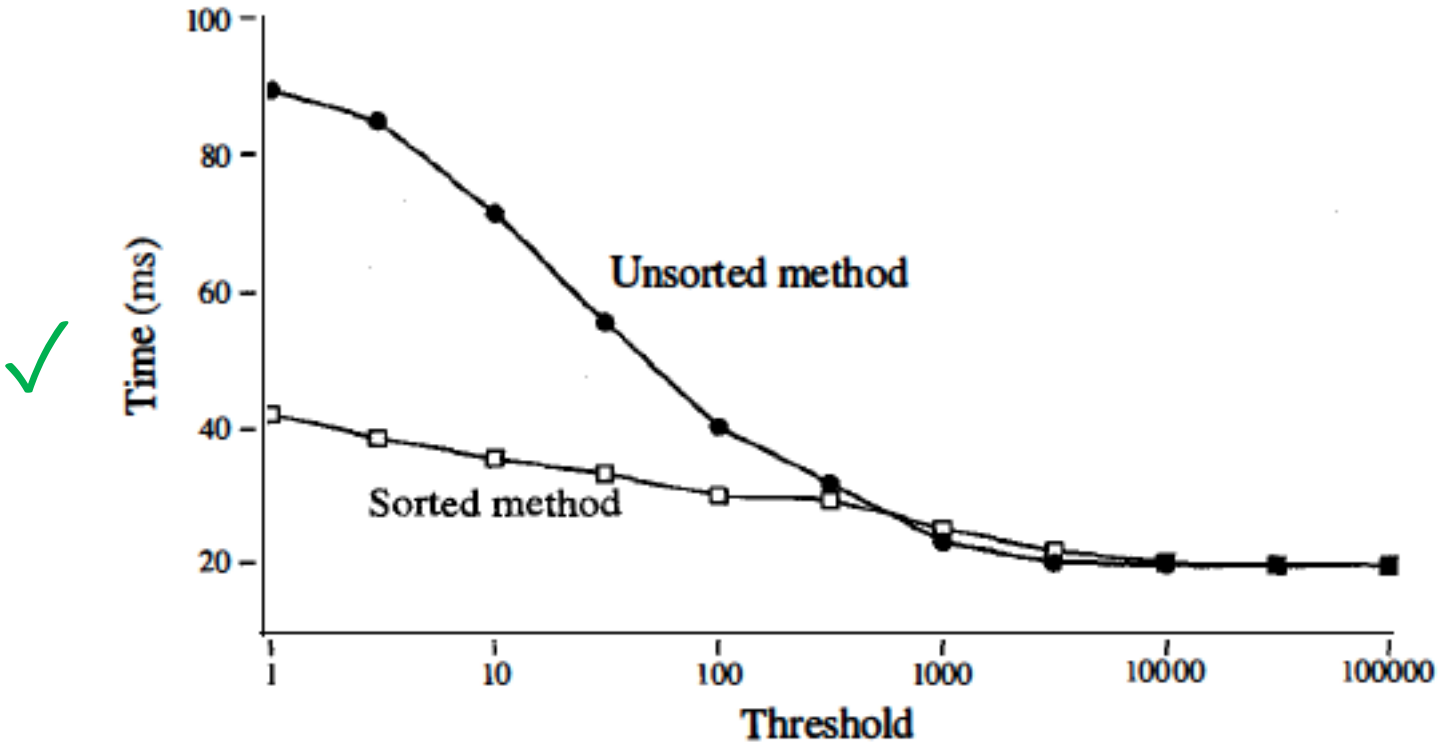
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Uso incorreto da escala em logaritmo



[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Uso correto da escala em logaritmo



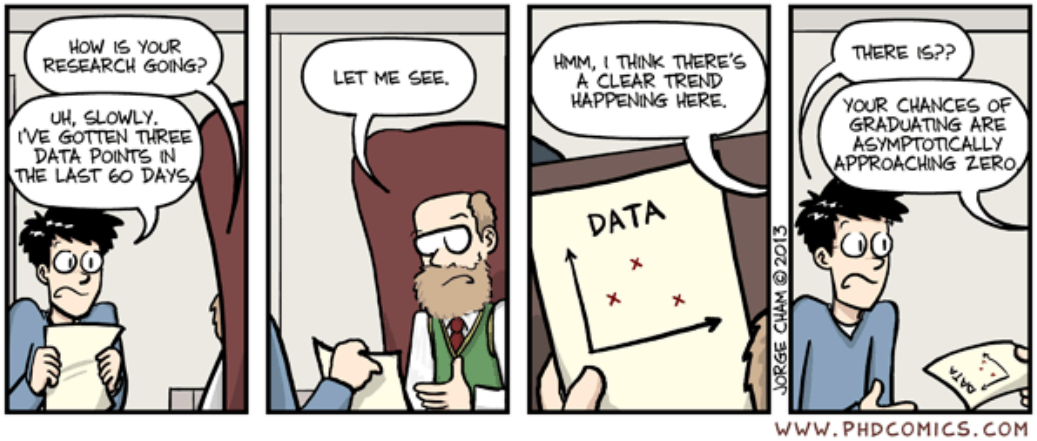
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Planejamento do gráfico

- O eixo x é para parâmetros que são variados
  - var. independente
- O eixo y é para variáveis que são respostas aos parâmetros
  - var. dependente
- Use marcadores de círculo, triângulos, quadrados
  - Traços e cruzes são aceitáveis, caso sejam fáceis de visualizar
- Seja consistente no uso dos tons
  - Gráficos profissionais usam preto e branco
  - Se for usar cor, seja consistente no uso das cores
- Utilize ao máximo a área para apresentar os dados
- Evite decorações, grades, quadros e outros ornamentos
- As legendas não devem se misturar com os dados
- Atenção no uso de escala em logaritmo
  - Se um algoritmo for 30% melhor que outro, o seu desempenho fica parecido com os demais em uma escala log-log

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Gráficos avançados



Figuras em 3D

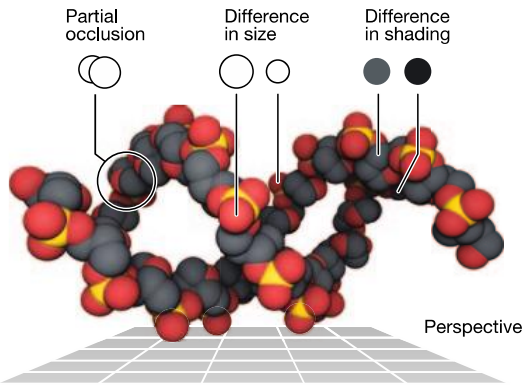


Figure 1 | Space-filling model of the DNA backbone. Depth cues enable us to perceive two-dimensional images as three-dimensional objects.

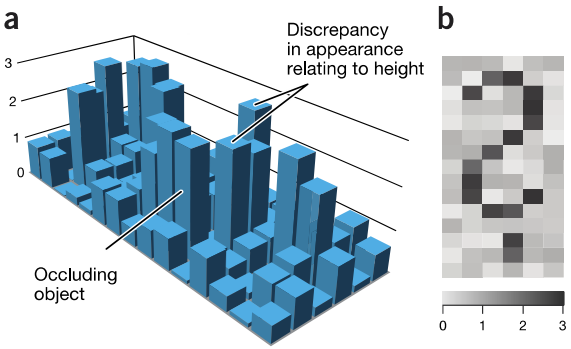


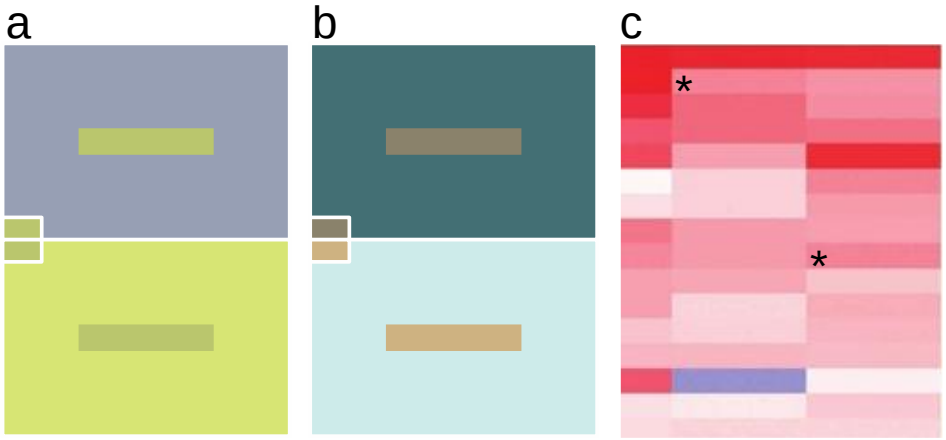
Figure 2 | Three-dimensional representation of abstract data. (**a**) Data occlusion and interference of visual encodings with depth cues can be problematic in three-dimensional space. (**b**) The same data as in **a** plotted as a two-dimensional heat map.

- Projeção 3D só deve ser usada em objetos espaciais, sendo que há efeitos colaterais na projeção 2D
- Adequado para ferramentas que possam funcionalidade de girar a figura

- Visualização 3D pode esconder outras partes da figura
- O uso de escala de cores em um gráfico 2D é melhor

[1] N. Gehlenborg and B. Wong, 2012, Points of view: Into the third dimension, *Nature Methods*, v. 9, n. 9, p. 851.

Percepção das cores



Percepção das cores pode variar
A mesma cor pode parecer diferente (a) e (c)
Cores diferentes podem parecer iguais (b)

[1] B. Wong, 2011, Points of view: Avoiding color, Nature Methods, v. 8, n. 7, p. 525.

Escolhendo cores

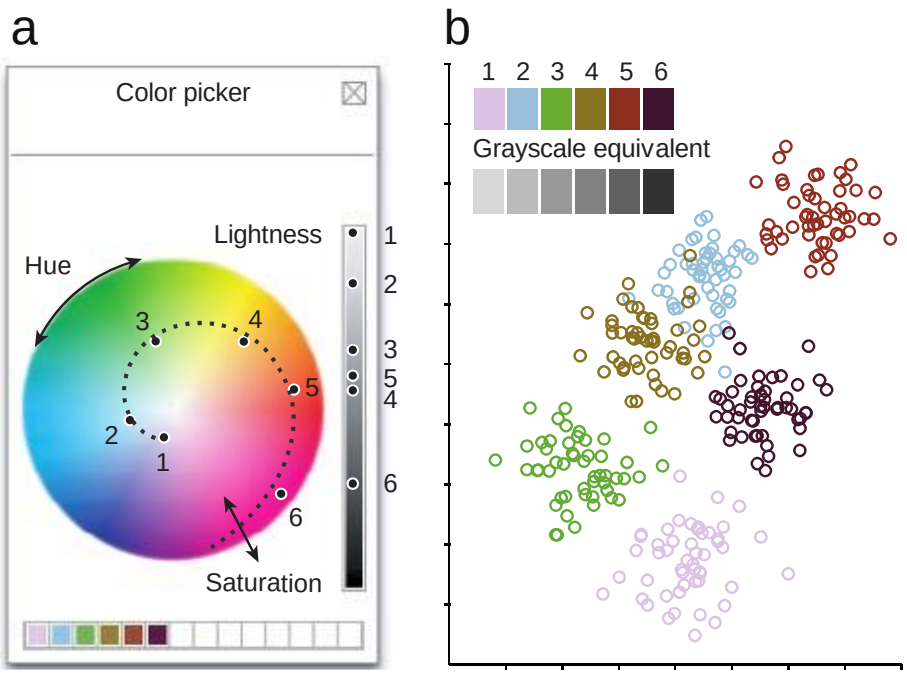


Figure 2 | Color has hue, saturation and brightness. (**a**,**b**) Colors can be tuned using a color picker (**a**). Spiraling through hue and saturation while varying lightness can generate a discernible color set distinguishable even in grayscale (points labeled 1–6).

Cores que se mantem em uma combinação espiral de matiz e saturação enquanto variam a luminosidade produzem cores mais diferenciadas, inclusive em tons de cinza

[1] B. Wong, 2010, Points of view: Color coding, Nature Methods, v. 7, n. 8, p. 573.
ColorBrewer (http://colorbrewer2.org/) and NASA's Color Tool (http://colorusage.arc. nasa.gov/ColorTool.php#1).

Eixos, marcas e grades

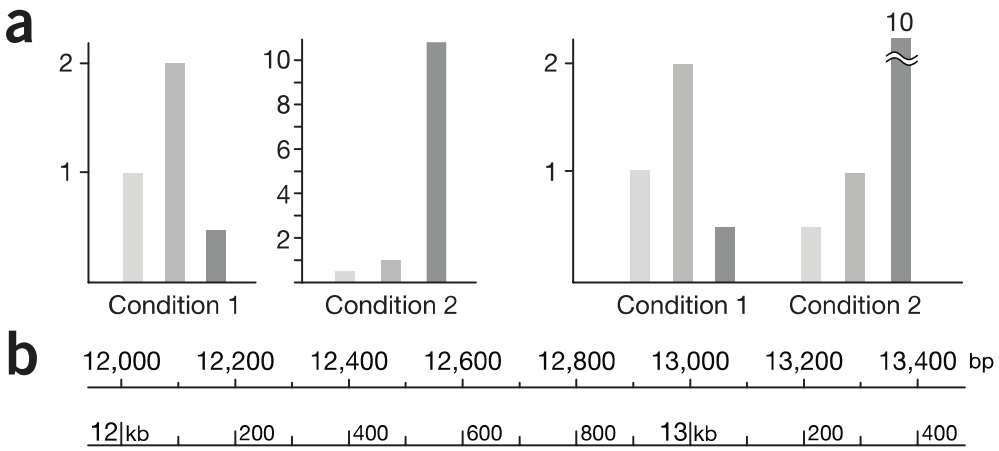


Figure 2 | Avoid unnecessary variation and repetition in axes and ticks. (**a**) If absolute differences are important, maintain axis scaling across panels. Draw a single *y* axis to emphasize that the scale is fixed. In bar plots, use breaks to shorten outlier elements that would otherwise compress the dynamic range of the data. (**b**) Duplication of nonsignificant digits in tick marks should be reduced or removed altogether by adjusting the units.

[1] M. Krzywinski, 2013, Points of view: Axes, ticks and grids, *Nature Methods*, v. 10, n. 3, p. 183.

Eixos, marcas e grades

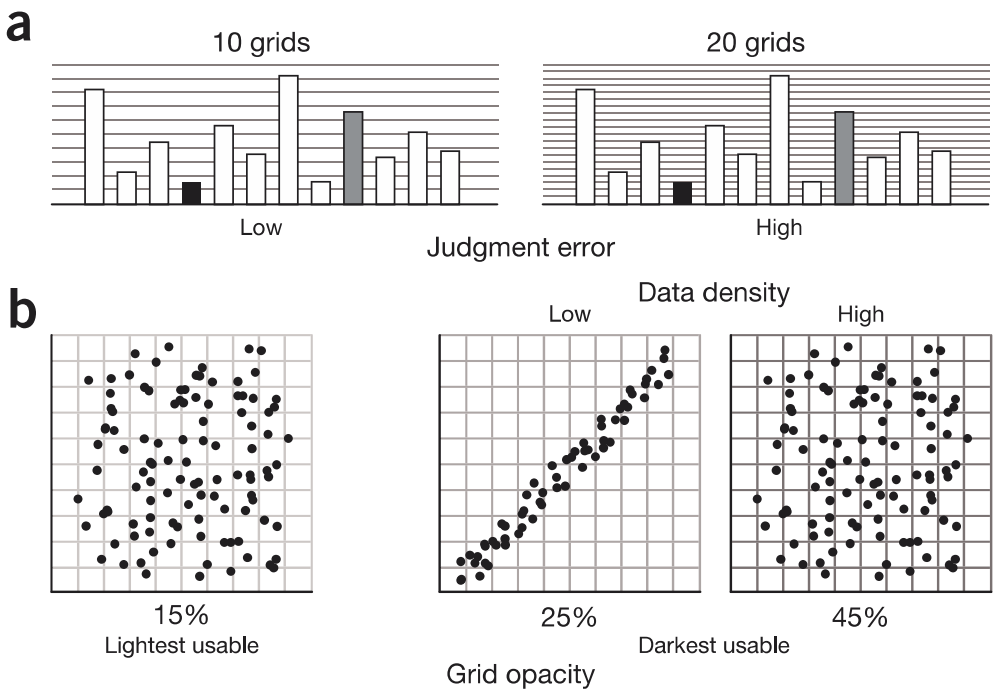


Figure 3 | Control grid density and transparency to maintain separability from data and other grids. (**a**) User studies in which readers were asked to assess the proportion of bar heights show that the density of grid lines is correlated with judgment error[4]. (**b**) Grid line transparency is most effective in the range of 15%–45%, depending on data density[4].

[1] M. Krzywinski, 2013, Points of view: Axes, ticks and grids, *Nature Methods*, v. 10, n. 3, p. 183.

Eixos, marcas e grades

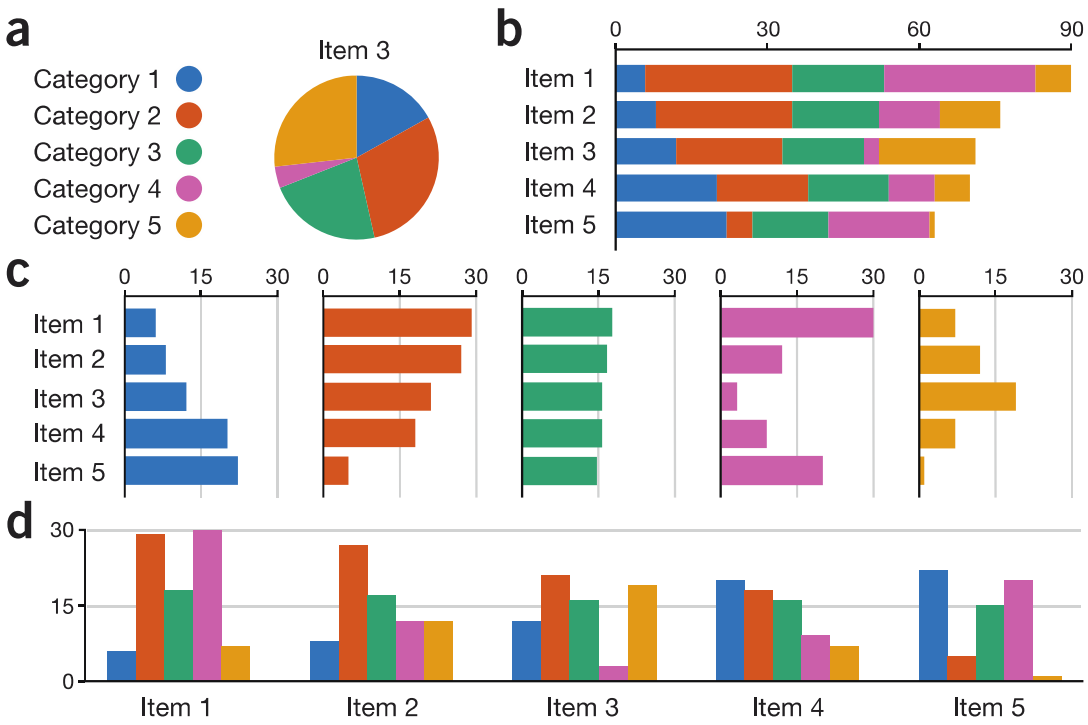


Figure 1 | Variants of bar charts and a pie chart encoding the same data.
(**a**) Values in different categories are difficult to compare in pie charts.
(**b**) Stacked bar charts enable comparison of overall values across items.
(**c**) Layered bar charts support comparison of values within categories.
(**d**) Grouped bar charts allow comparison of values across categories.

[1] M. Streit and N. Gehlenborg, 2014, Points of View: Bar charts and box plots, *Nature Methods*, v. 11, n. 2, p. 117.

Eixos, marcas e grades

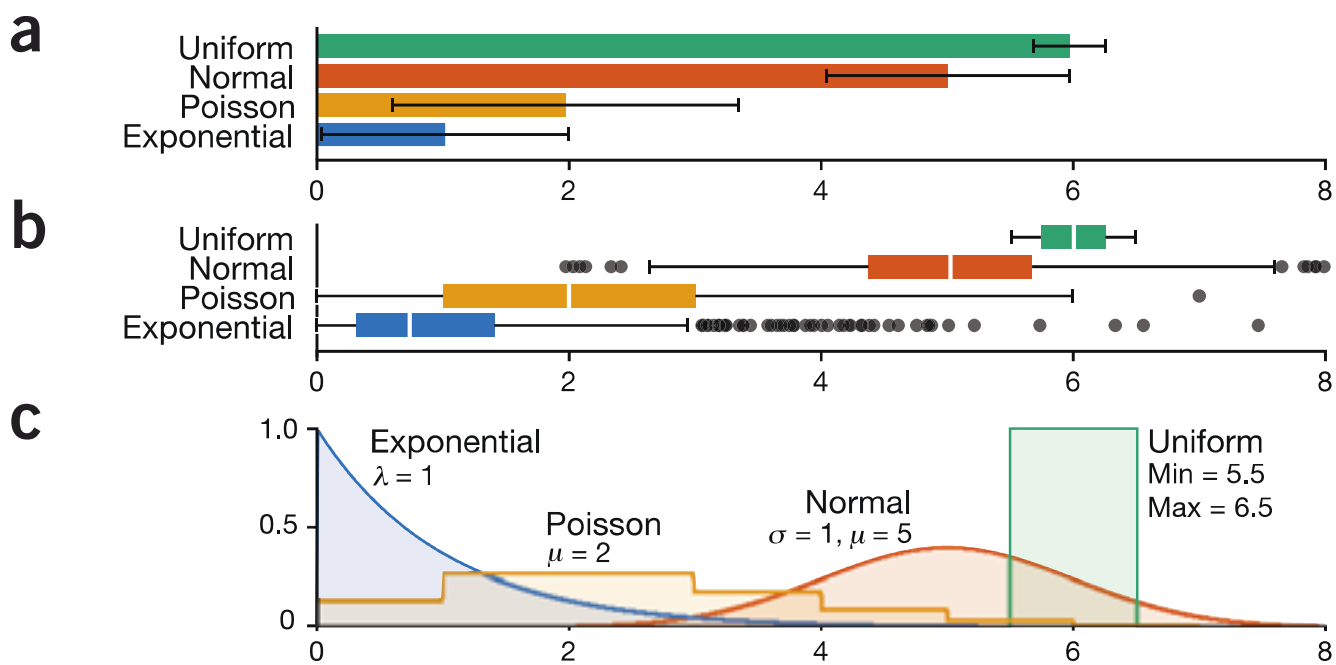


Figure 2 | Representation of four distributions with bar charts and box plots. (**a**) Bar chart showing sample means (n = 1,000) with standard-deviation error bars. (**b**) Box plot (n = 1,000) with whiskers extending to ±1.5 × IQR. (**c**) Probability density functions of the distributions in **a** and **b**. λ, rate; μ, mean; σ, standard deviation.

[1] M. Streit and N. Gehlenborg, 2014, Points of View: Bar charts and box plots, *Nature Methods*, v. 11, n. 2, p. 117.

Dados temporais

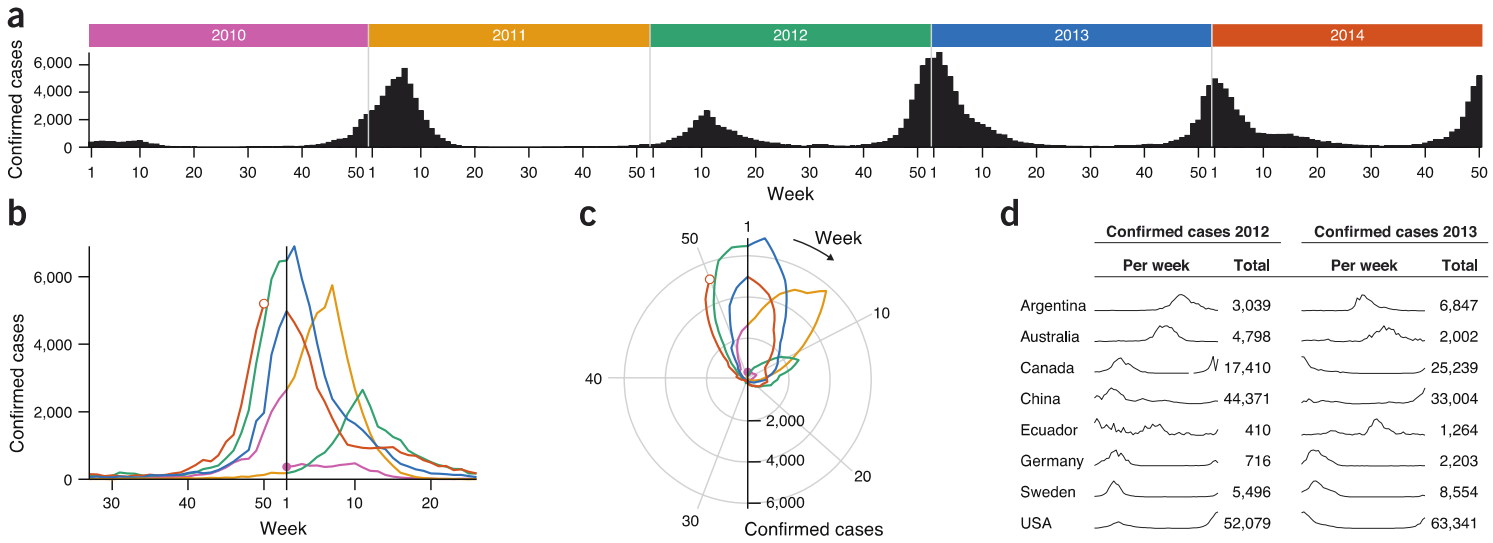


Figure 1 | Alternative representations of confirmed cases of influenza types A and B. (**a**) Bar chart plotting confirmed cases in the United States from week 1 in 2010 to week 50 in 2014. (**b**) Superimposed line chart showing the data from **a** as individual curves. The position of week 1 on the horizontal axis was chosen to emphasize the annual peak. Start and end point of the time series are indicated by filled and hollow circles, respectively. (**c**) Radar chart showing the data from **a** as a single continuous curve. (**d**) Sparklines depicting the overall influenza patterns of different countries for the years 2012 and 2013.

[1] M. Streit and N. Gehlenborg, 2015, Points of view: Temporal data, *Nature Methods*, v. 12, n. 2, p. 97.

Diagramas

- São muito usados em computação
  - Processos
  - Arquiteturas
  - Estruturas de dados
  - Algoritmos
  - Relacionamentos
  - Interfaces
- Em linhas gerais os diagramas tem como propósito apresentar: estruturas, processos, relacionamentos ou estados
  - É um erro comum tentar misturar, em um único diagrama, esses propósitos

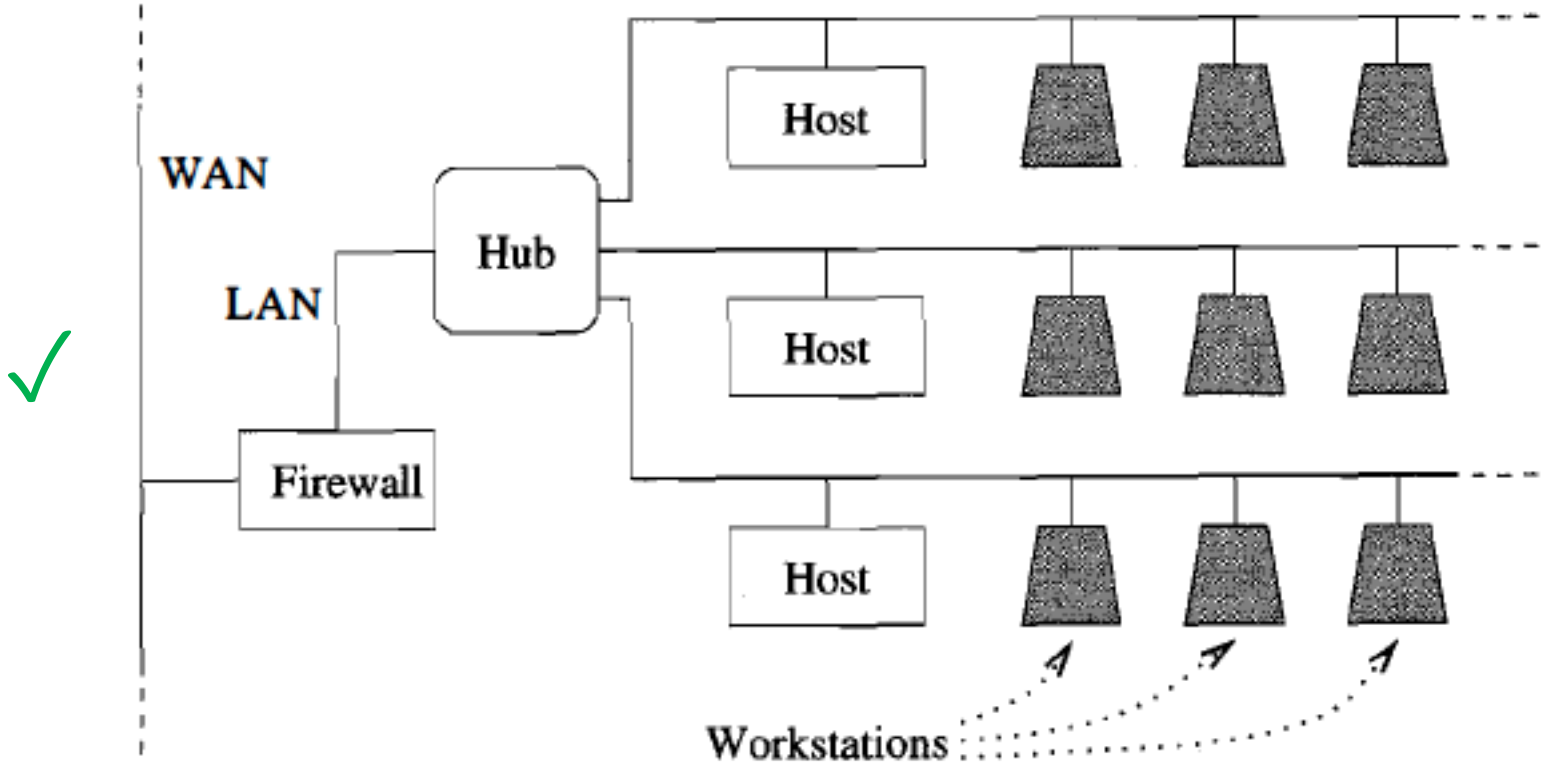
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Formatação de diagramas

- Evite deixar os diagramas escuros
  - Deixo-o com o máximo de preenchimento em branco
- O diagrama é uma abstração
  - Não é necessário exagerar nos detalhes de implementação
- Todos os itens do diagrama devem ser explicados no texto
  - Coloque apenas o essencial para o seu bom entendimento
- Como nos gráficos, procure trabalhar em tons de cinza
- Se usar cores, use-as de modo consistente
- Mantenha a identidade visual do artigo

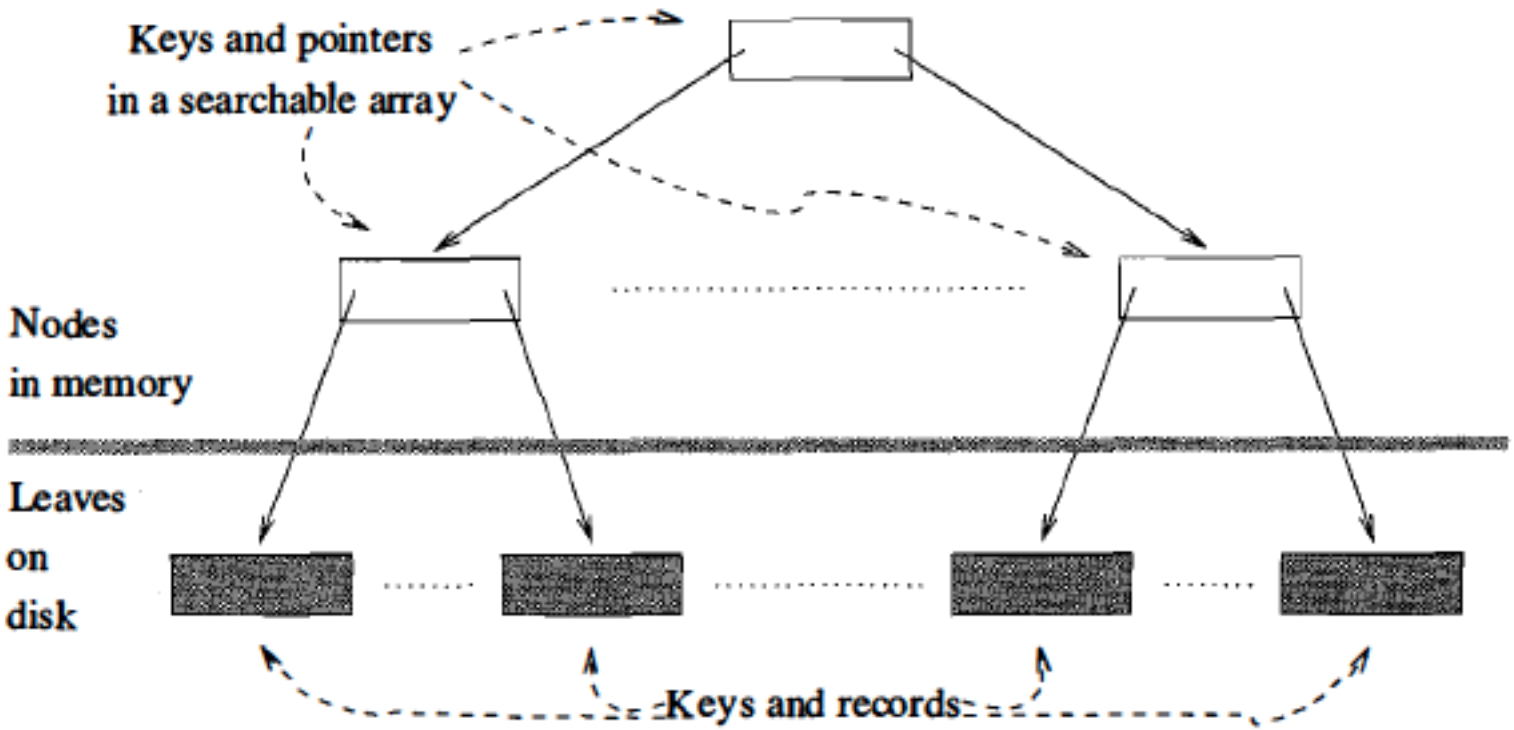
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Exemplo de arquitetura - firewall



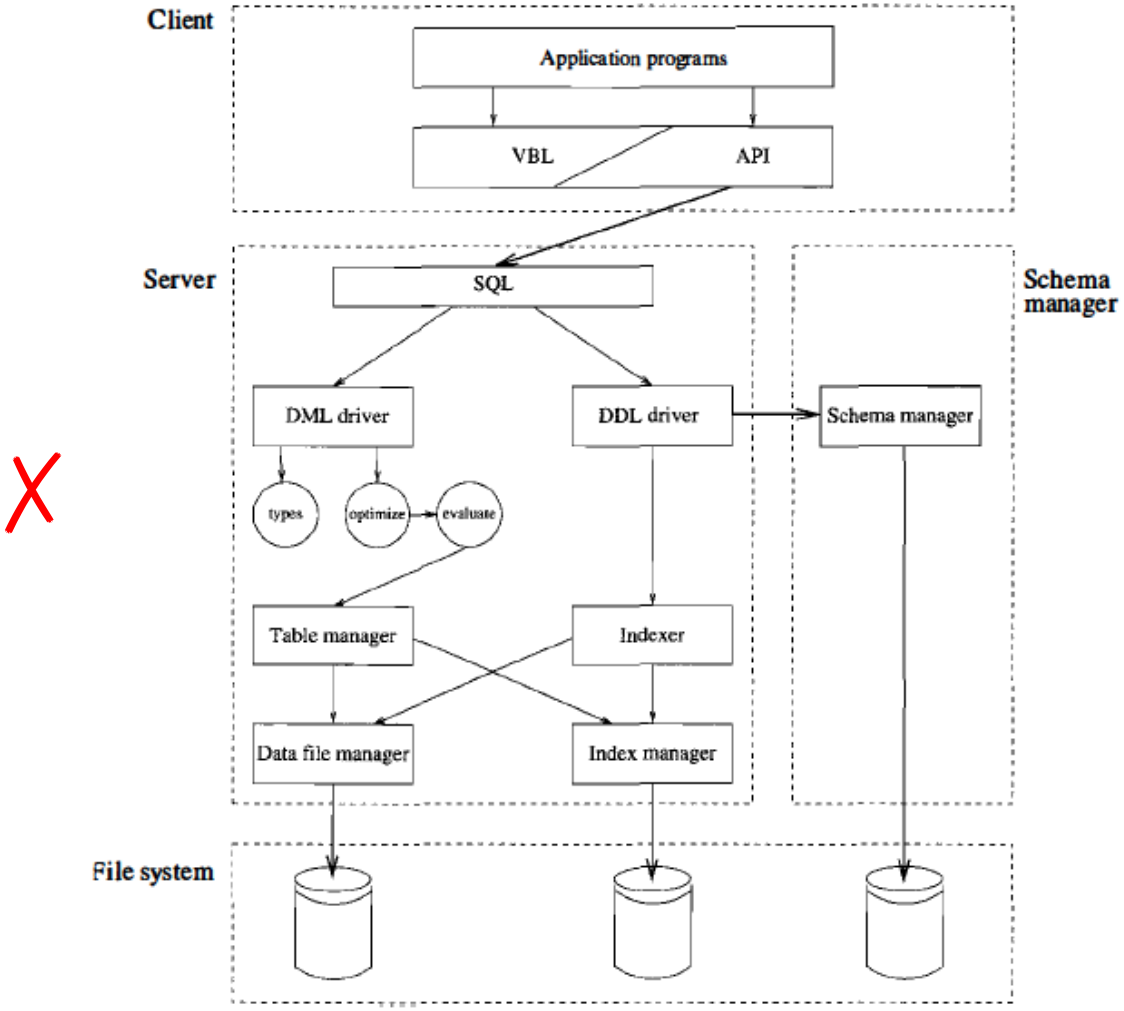
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Exemplo de estrutura de dados - árvore



[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

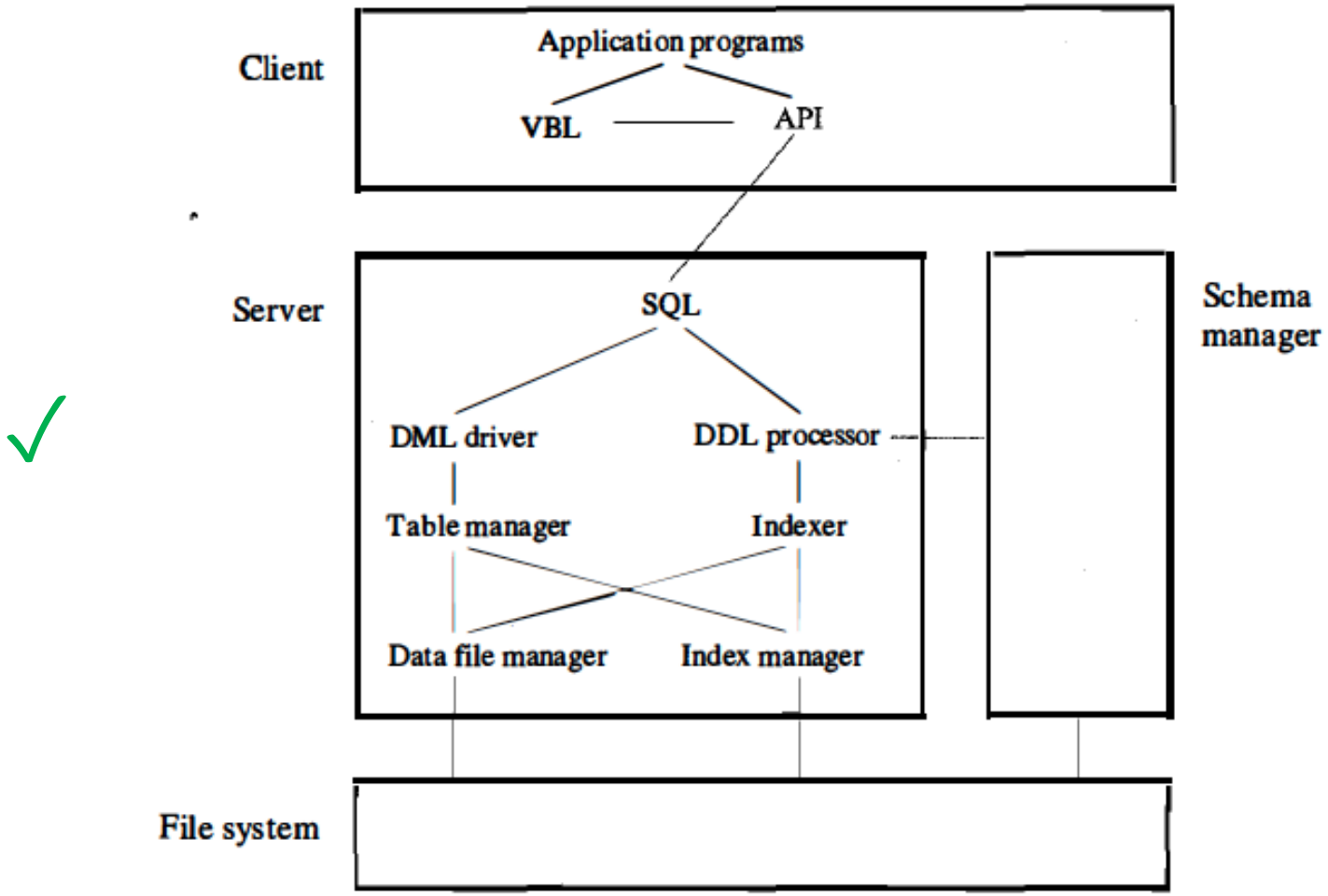
Arquitetura com problemas no nível de detalhamento



- Fonte pequena
- Linhas muito finas
- Algumas partes internas devem ser omitidas
- Setas trazem pouca informação
- Algumas deveriam ser bi direcionadas

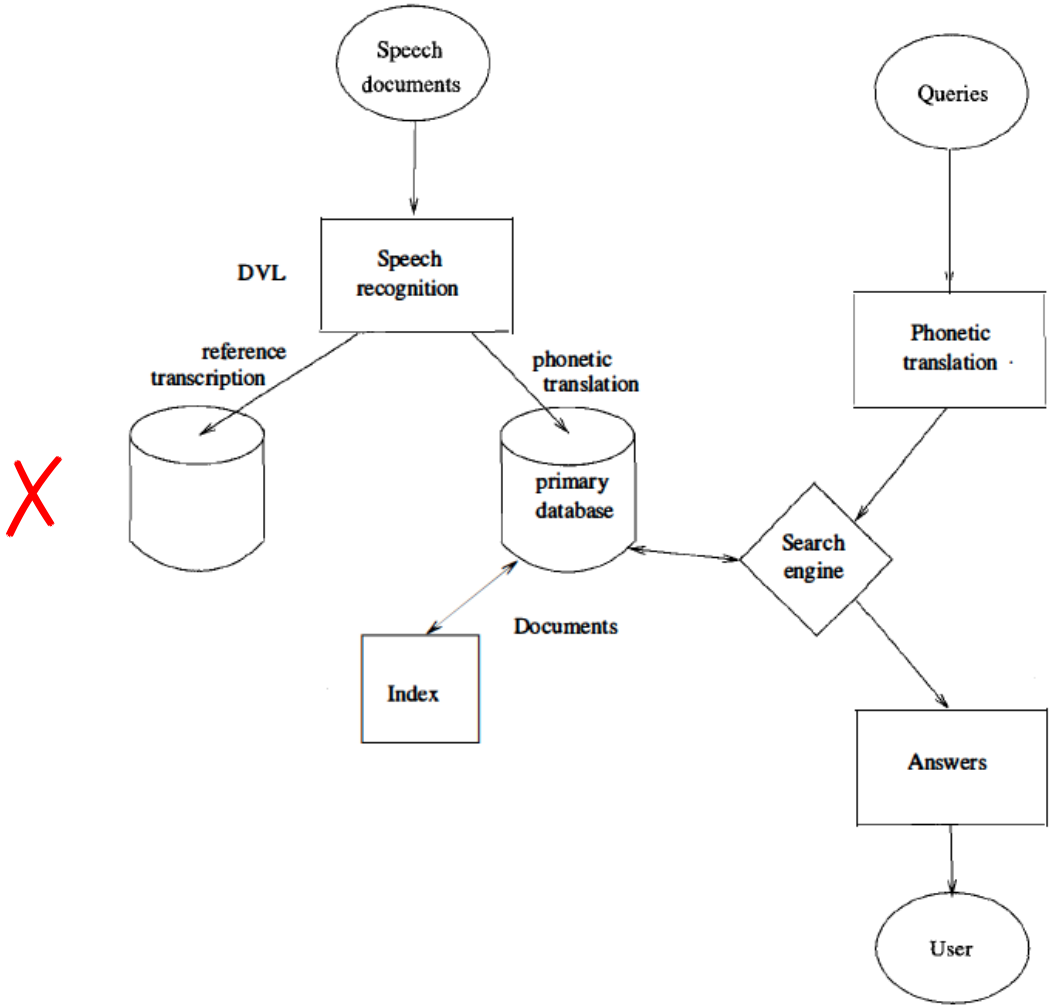
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Arquitetura com nível de detalhamento adequado



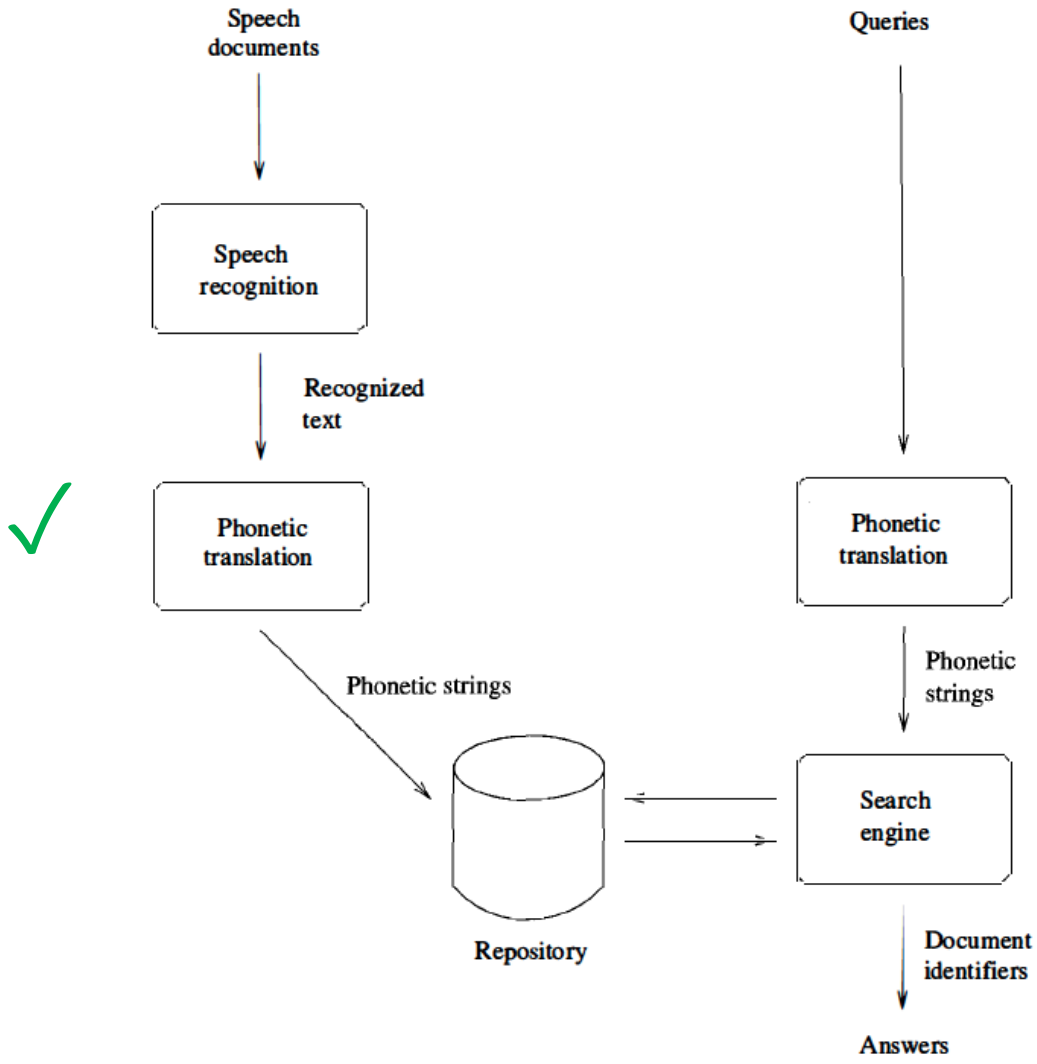
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Diagrama de processo



[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Diagrama de processo



[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Uso de setas

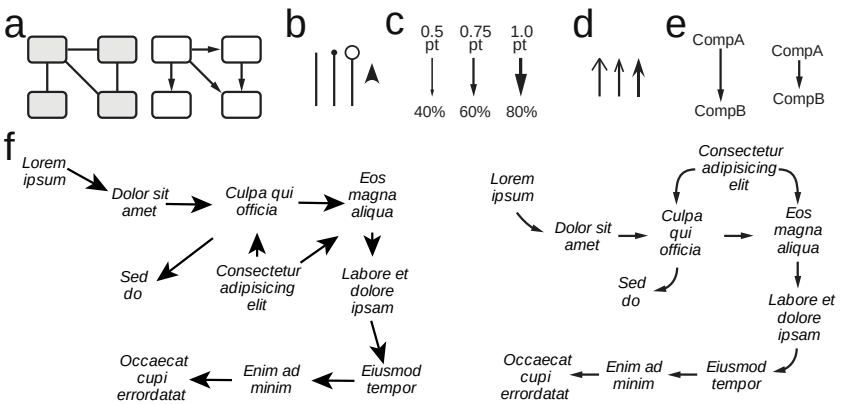


Figure 2 | Functional qualities of arrows. (**a**) The use of arrows versus lines as connectors suggests a certain functional relationship. (**b**) Alternatives to arrows as leader lines. (**c**) Reasonably sized arrows clearly indicate direction without being a distraction. (**d**) Trapped whitespace in 'open' arrowheads creates optical illusions that can attract unwanted attention. (**e**) Whitespace at the ends of the arrows makes them easy to discriminate from other content. (**f**) Orienting arrows in similar directions creates natural visual flow.

[1] B. Wong, 2011, Points of view: Arrows, *Nature Methods*, v. 8, n. 9, p. 701.

Tabelas

- Tabelas são usadas quando não é adequado apresentar figuras ou gráficos
- Tabelas bem projetadas tem uma organização lógica hierarquizada
- As colunas tem um cabeçalho com os nomes dos atributos
- Tabelas devem ser abertas e livres, com bastante espaço em branco
  - Devem ter poucas linhas horizontais ou verticais
  - Não devem ser densas

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Comparação de parâmetros

STATISTICS	SMALL	LARGE
Characters	18,621	1,231,109
Words	2,060	173,145
After stopping	1,200	98,234
Index size	1.31 Kb	109.0 Kb

	Collection	
	Small	Large
File size (Kb)	18.2	1,202.3
Index size (Kb)	1.3	109.0
Number of words	2,060	173,145
After stopping	1,200	98,234

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Bom uso da hierarquização

Parameter	Data set			
	SINGLE		MULTIPLE	
	CPU (msec)	Effective (%)	CPU (msec)	Effective (%)
n ($k = 10$, $p = 100$)				
2	57.5	55.5	174.2	22.2
3	21.5	50.4	79.4	19.9
4	16.9	47.5	66.1	16.3
k ($n = 2$, $p = 100$)				
10	57.5	51.3	171.4	21.7
100	60.0	56.1	163.1	21.3
1000	111.3	55.9	228.8	21.4
p ($n = 2$, $k = 10$)				
100	3.3	5.5	6.1	1.2
1000	13.8	12.6	19.8	2.1
10,000	84.5	56.0	126.4	6.3
100,000	—	—	290.7	21.9

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Excesso de informação

Pass	Output	Size Mb	Size %	CPU Hr:Min	Mem Mb
Pass 1:					
Compression	Model	4.2	0.2	2:37	25.6
Inversion	Vocabulary	6.4	0.3	3:02	18.7
Overhead				0:19	2.5
Total		10.6	0.5	5:58	46.8
Pass 2:					
Compression	Text	605.1	29.4	3:27	25.6
	Doc. map	2.8	0.1		
Inversion	Index	132.2	6.4	5:25	162.1
	Index map	2.1	0.1		
	Doc. lens	2.8	0.1		
	Approx. lens	0.7	0.0		
Overhead				0:23	2.5
Total		745.8	36.3	9:15	190.2
Overall		756.4	36.8	15:13	190.2

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

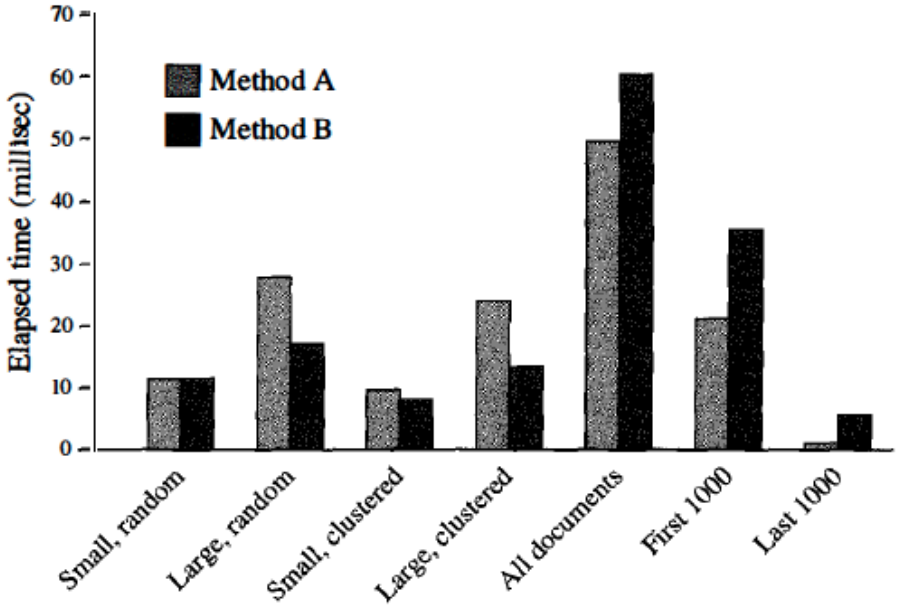
Simplificação

Task	Size (Mb)	CPU (Hr:Min)	Memory (Mb)
Pass 1:			
Compression	4.2	2:37	25.6
Inversion	6.4	3:02	18.7
Overhead	—	0:19	2.5
Total	10.6	5:58	46.8
Pass 2:			
Compression	607.9	3:27	25.6
Inversion	137.8	5:25	162.1
Overhead	—	0:23	2.5
Total	745.8	9:15	190.2
Overall	756.4	15:13	190.2

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Comparação entre tabela e gráfico de barras

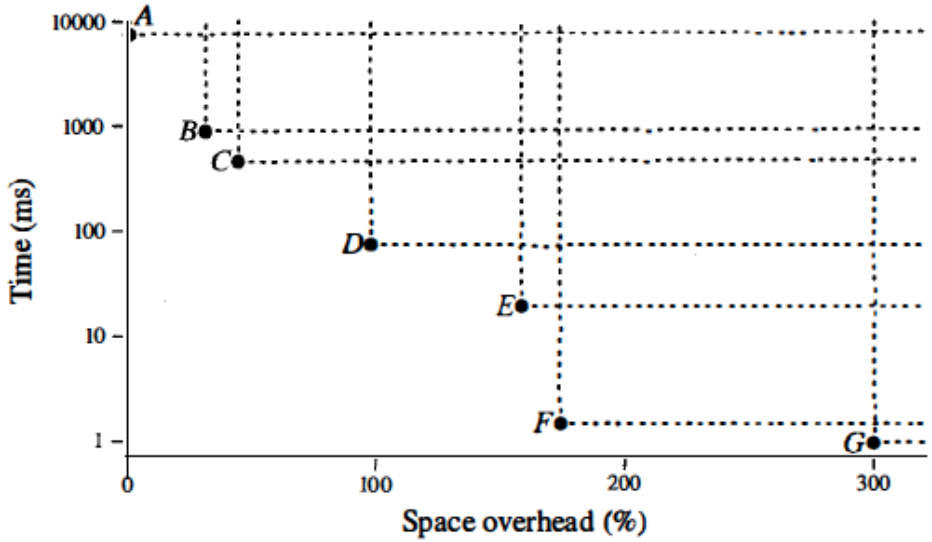
Data set	Method	
	A	B
Small, random	11.5	11.6
Large, random	27.9	17.1
Small, clustered	9.7	8.2
Large, clustered	24.0	13.5
All documents	49.4	60.1
First 1000	21.1	35.4
Last 1000	1.0	5.5



[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

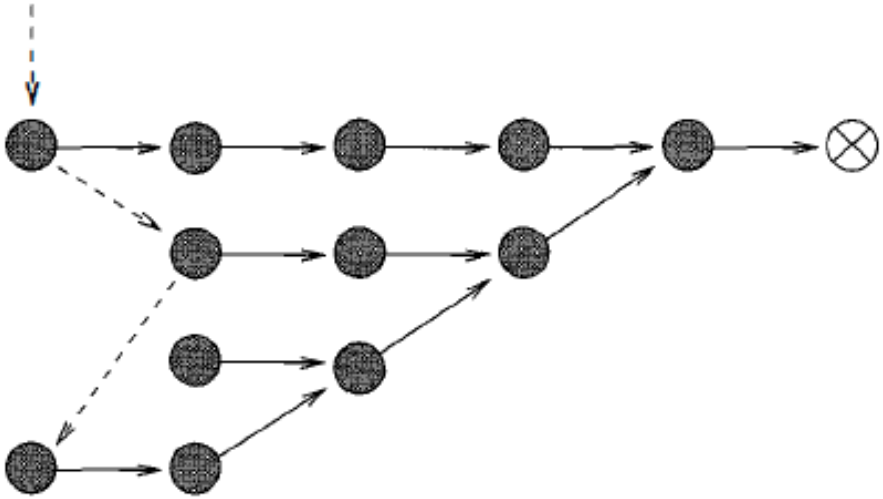
Comparação entre tabela e gráfico de barras

Method	Space (%)	Time (ms)
A	1.0	7,564.5
B	31.7	895.6
C	44.7	458.4
D	97.8	71.8
E	158.1	18.9
F	173.7	1.4
G	300.0	0.9

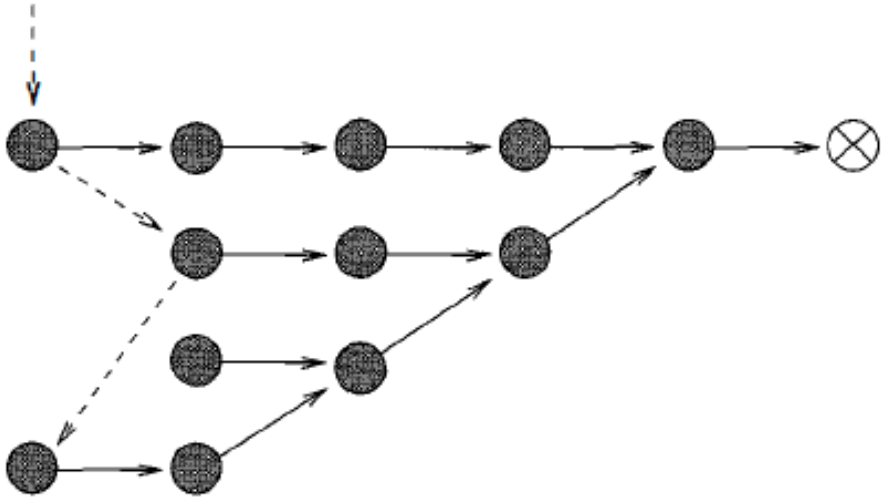


[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Legendas



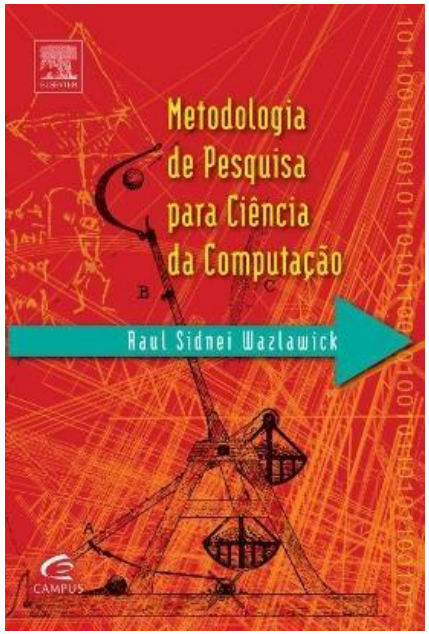
Estrutura direcionada

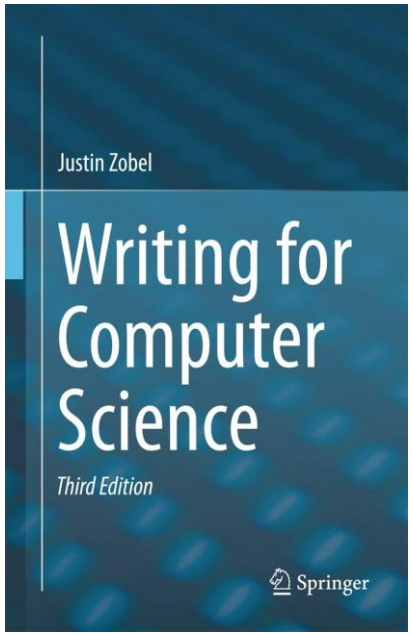


Estrutura direcionada de listas com uma cauda em comum. O nó cruzado é um sentinela. Linhas sólidas são ponteiros dentro da lista. Linhas pontilhadas comunicam as listas

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.

Referências





Leitura adicional Writing for Computer Science – Capítulo 11

Slides e vídeos em: https://eic.cefet-rj.br/~eogasawara/metodologia-cientifica/

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.
[2] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.